Pushing the boundaries: Automated Geometry Alignment with ‘BRDR' and ‘brdrQ’

A Python library aligning thematic geometries to reference borders

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FOSS4GE BE - Brussels

15/10/2026

geosquare

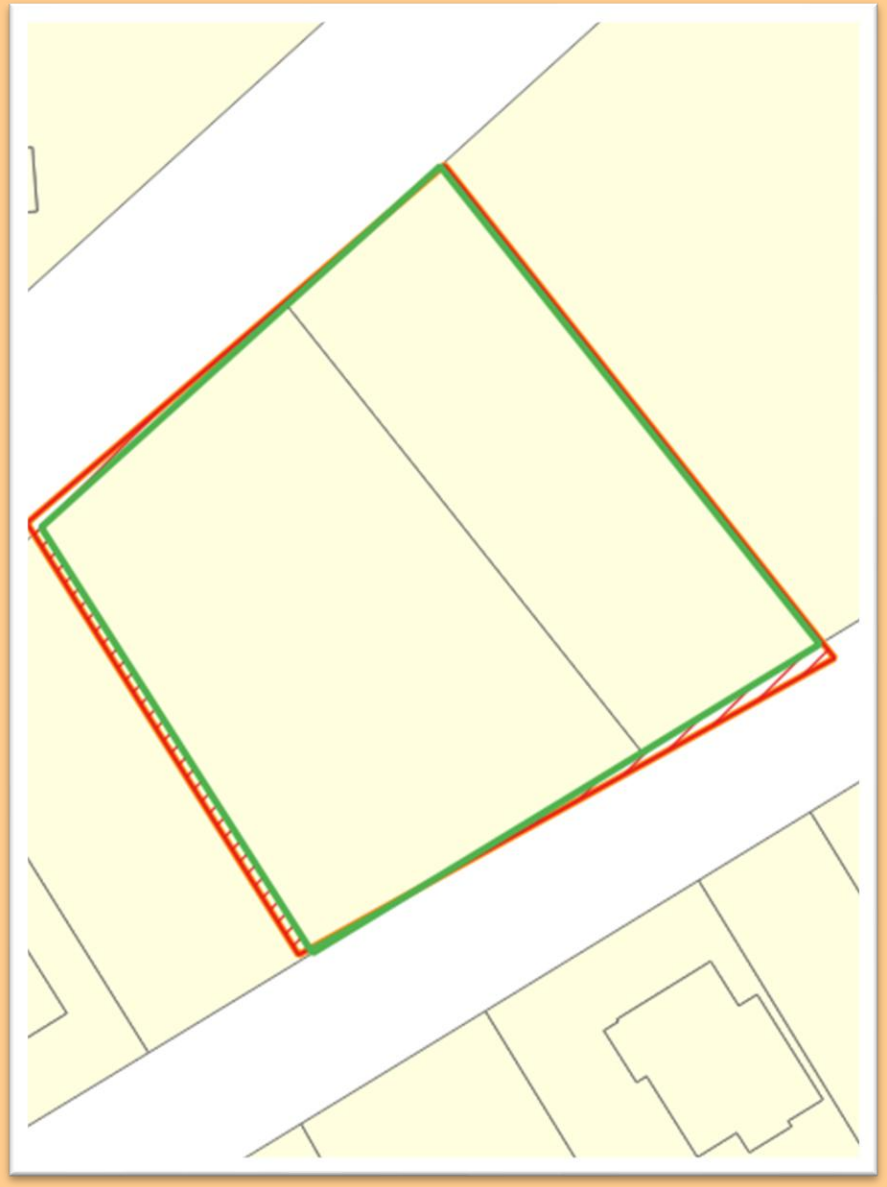
geosquare



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Agenda



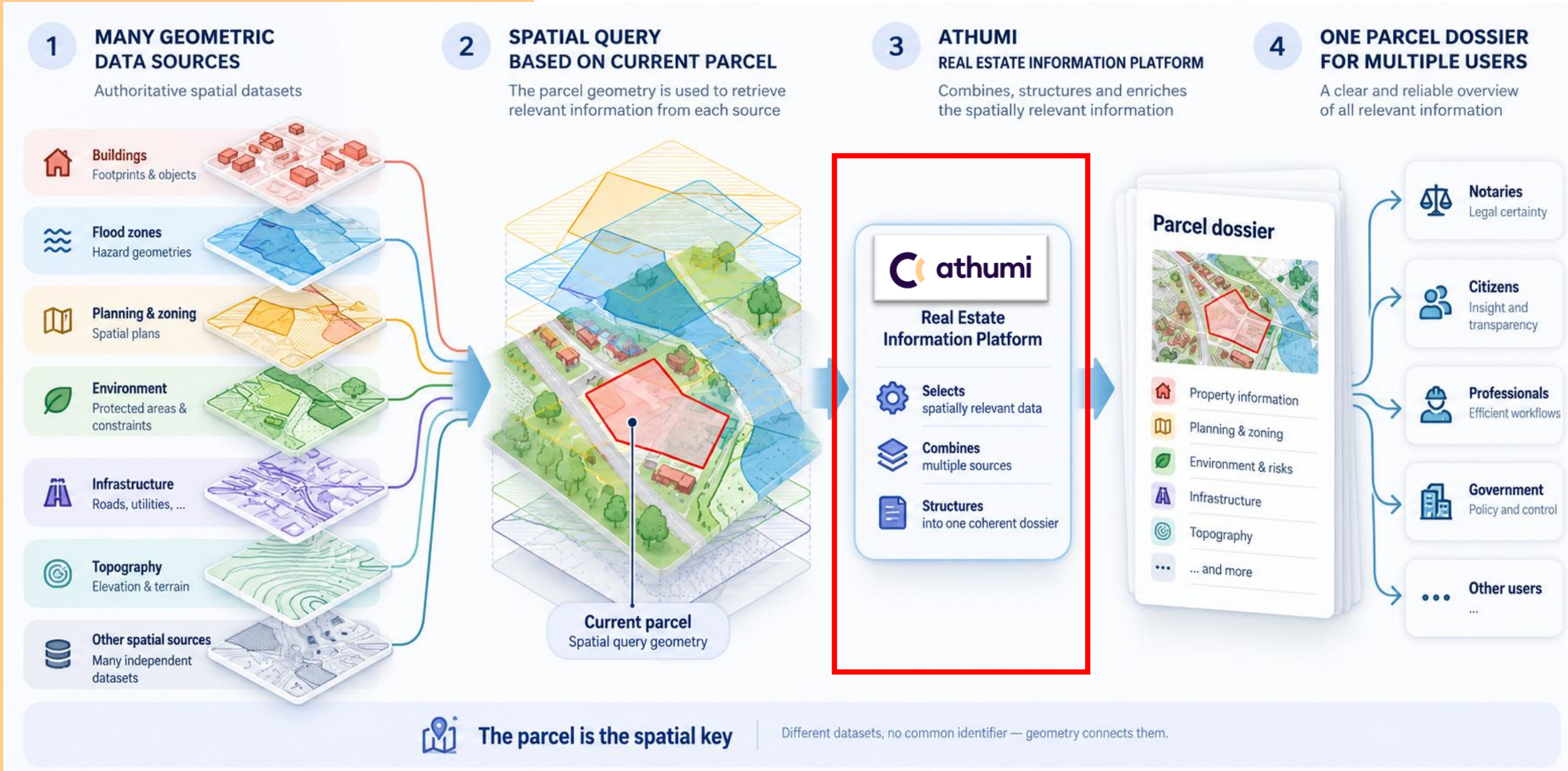
'BRDR' : Alignment of Geospatial Boundaries

- Why? The origin of the project/software
- What? BRDR & brdrQ
- How? The algorithm
- brdrQ Demo
- Use cases

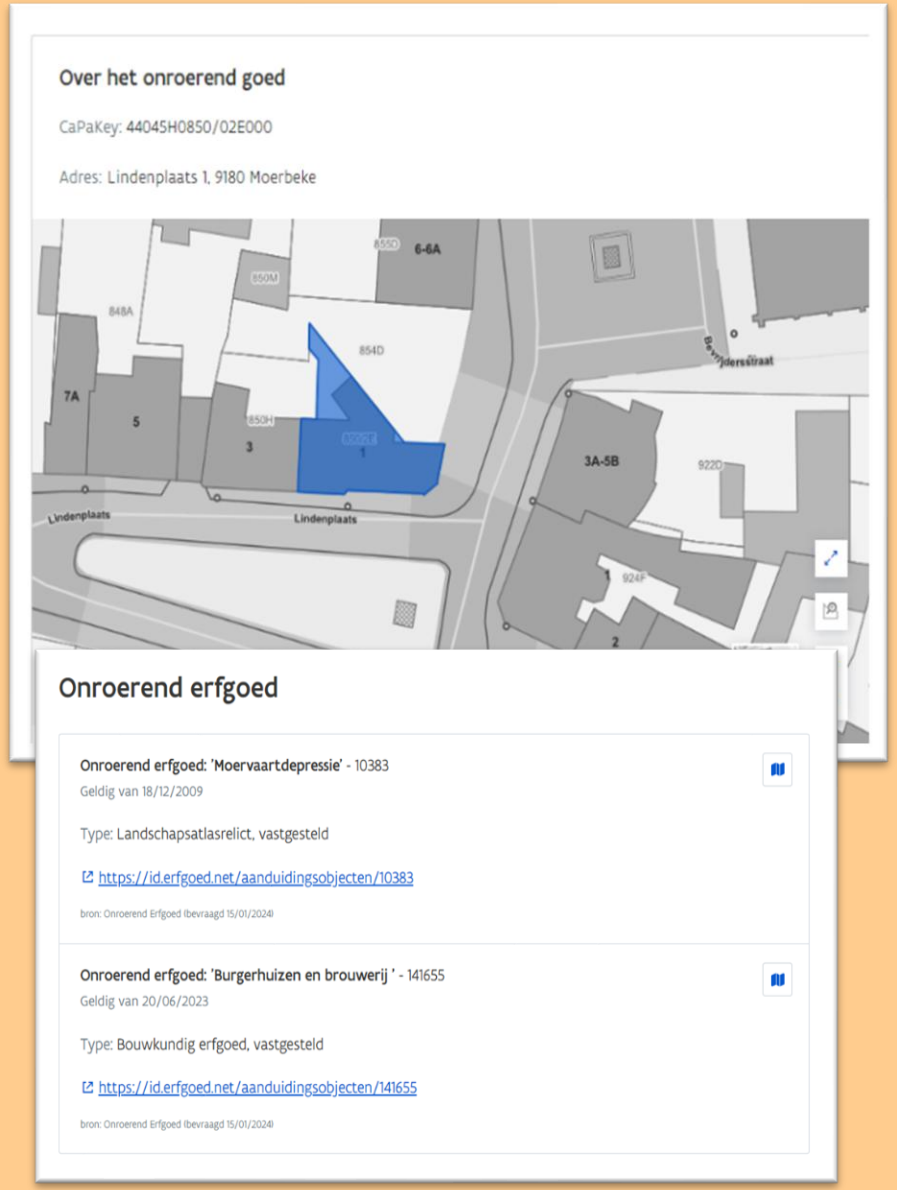
https://github.com/OnroerendErfgoed/brdr
https://github.com/OnroerendErfgoed/brdrQ

Why? The motive

athumi — *Flanders Data Utility Company*



Why? – Data quality!!



athumi

Flanders Data Utility Company

Exchanging **spatial** information

Where is protected heritage?

DATA SOURCES

Source X

……

Source: Flemish Heritage Agency

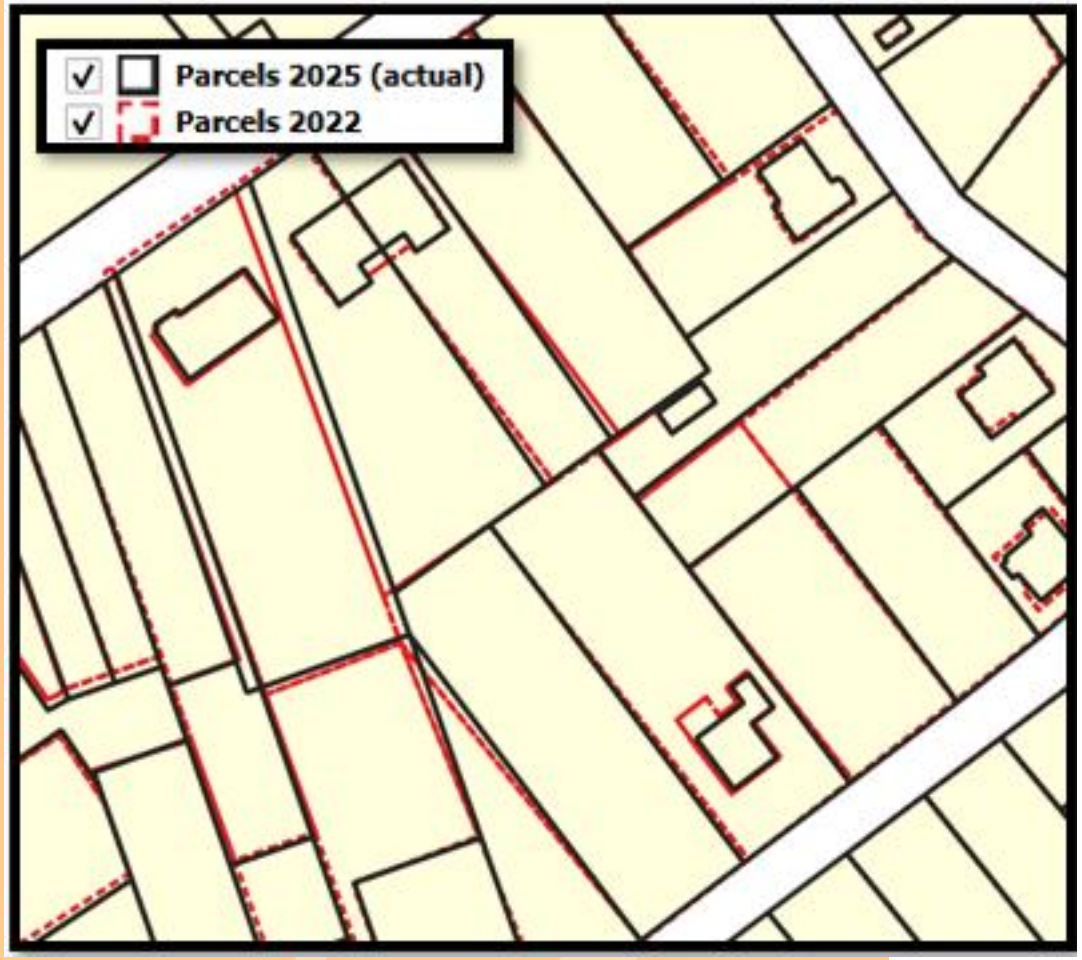
Vlaanderen is erfgoed

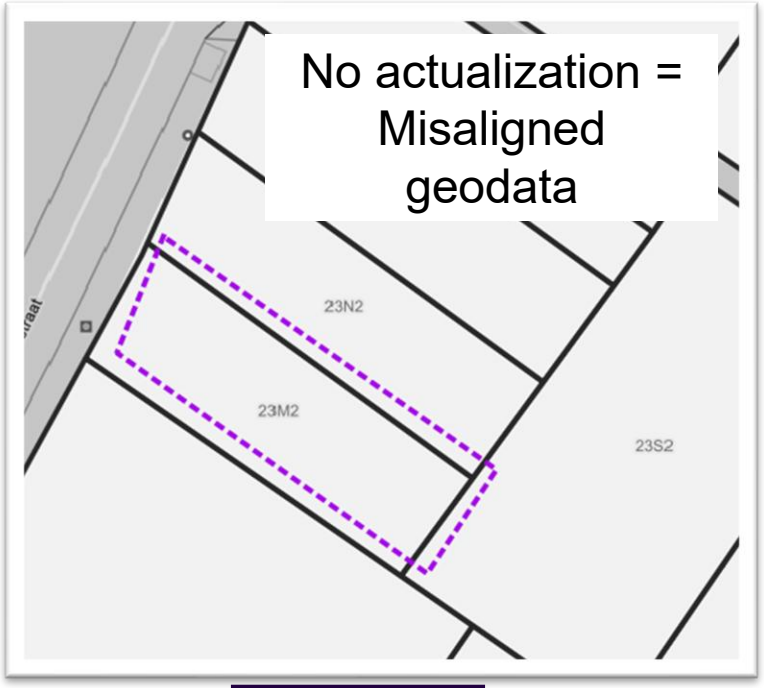
Source Z

……

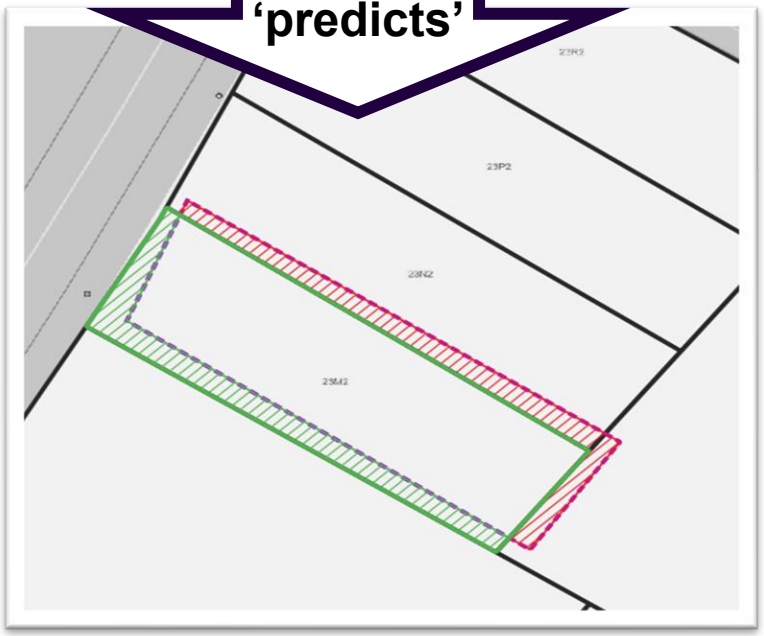
- Queries based on **geometries intersection**
- Geodata **quality** is crucial (!)
- **Thematic and reference layer's geometry diverge over time**
  - Splitting a parcel
  - Data quality improvements
  - …

Geodata quality diverging





BRDR ‘predicts’



What is BRDR?

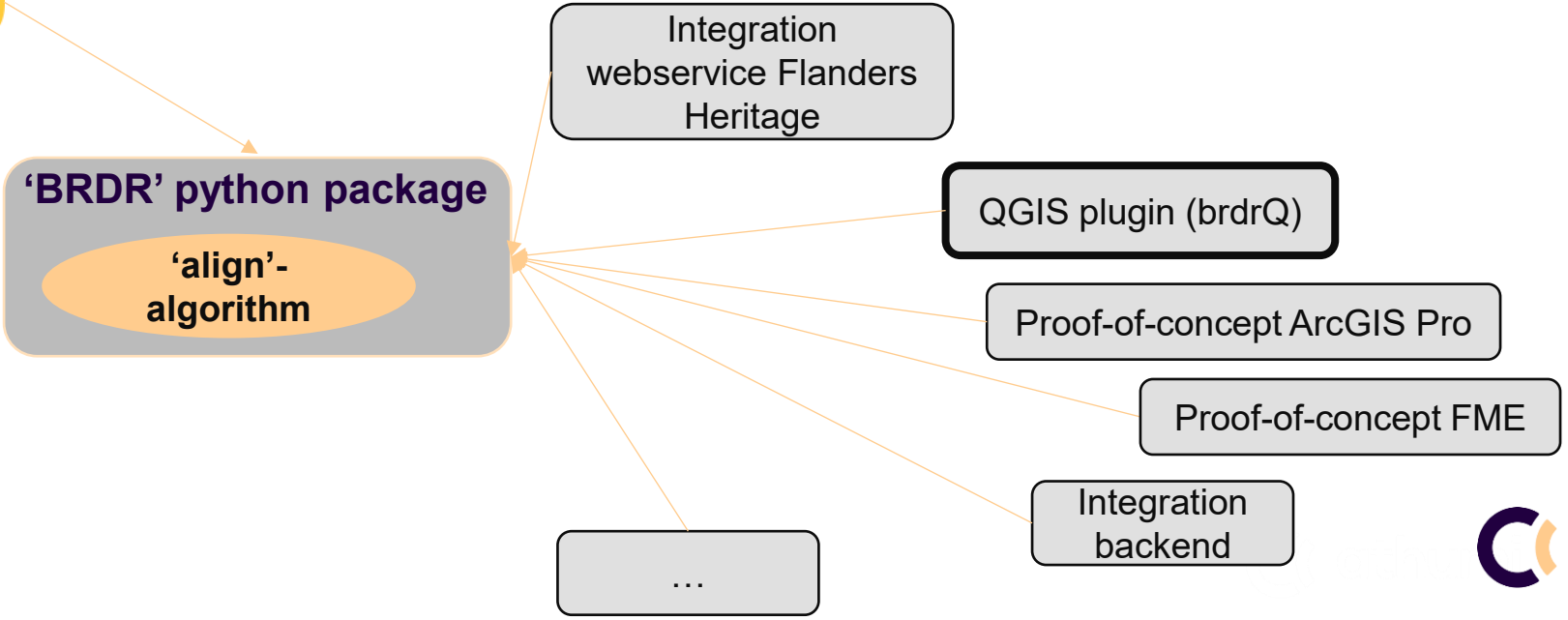
- **Open-source** (MIT-license)
- **Python package** (available on PyPi & GitHub)
- **Goal:** (re)aligning geometric borders to reference borders
- **Generic:**
  - Not tied to specific regions or organizations
  - Algorithm (calculation) separated from GUI/application integrations

```python
# CREATE AN ALIGNER
aligner = Aligner(crs="EPSG:31370")
# Load thematic WKT into Aligner
thematic_data = {
    "theme_id_1": geom_from_wkt("POLYGON ((170996.19647056088433601 170611.472792530228614
}
thematic_loader = DictLoader(thematic_data)
aligner.load_thematic_data(thematic_loader)

# Load reference data (GRB parcels)
reference_loader = GRBActualLoader(grb_type=GRBType.ADP, aligner=aligner)
aligner.load_reference_data(reference_loader)

# EXECUTE THE PREDICTOR
aligner.predictor()

#Get GeoJson-object of resulting prediction
featureclasses = aligner.get_results_as_geojson(resulttype=AlignerResultType.PREDICTIONS)
print (featureclasses['result'])
# SAVE PREDICTIONS TO file (GeoJSON)
aligner.save_results(path = 'output/',resulttype=AlignerResultType.PREDICTIONS)
```

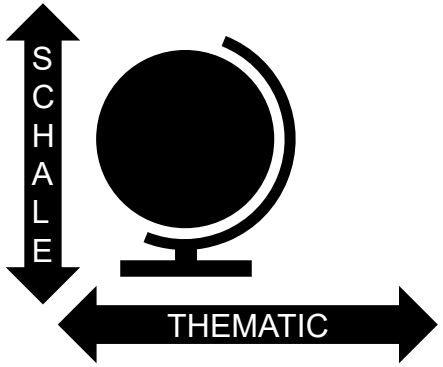


https://github.com/OnroerendErfgoed/brdr

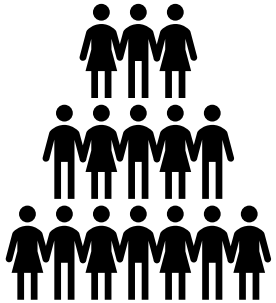
BRDR: OPEN & GENERIC



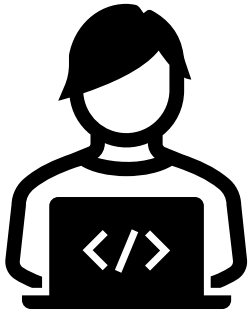
OPEN SOURCE



GENERIC USABLE



AVAILABLE FOR ALL



INTEGRATION

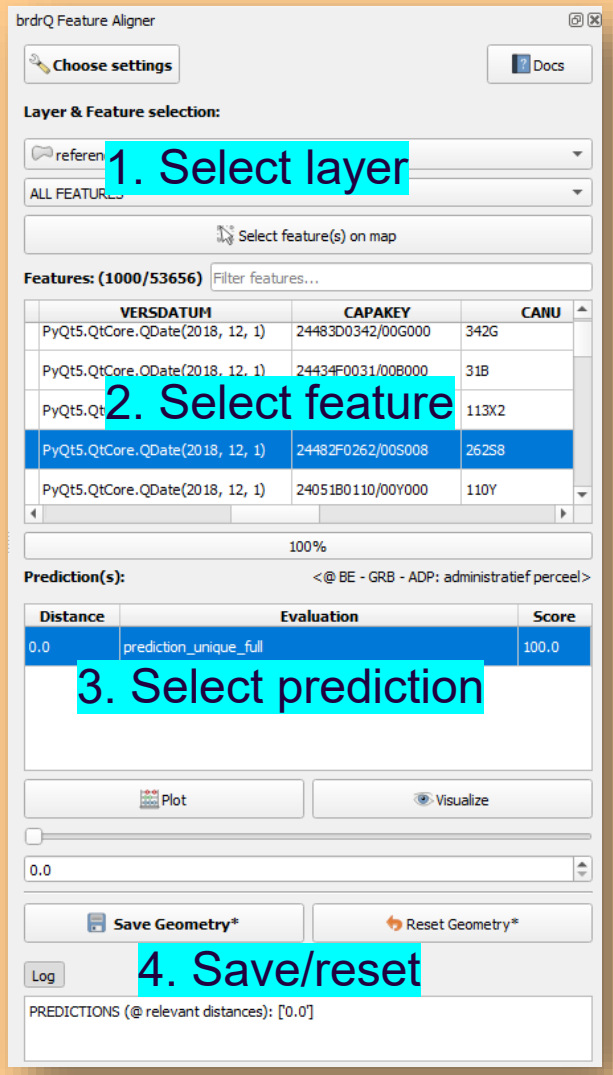
QGIS

ArcGIS Pro

FME

Web Service

What is brdrQ?

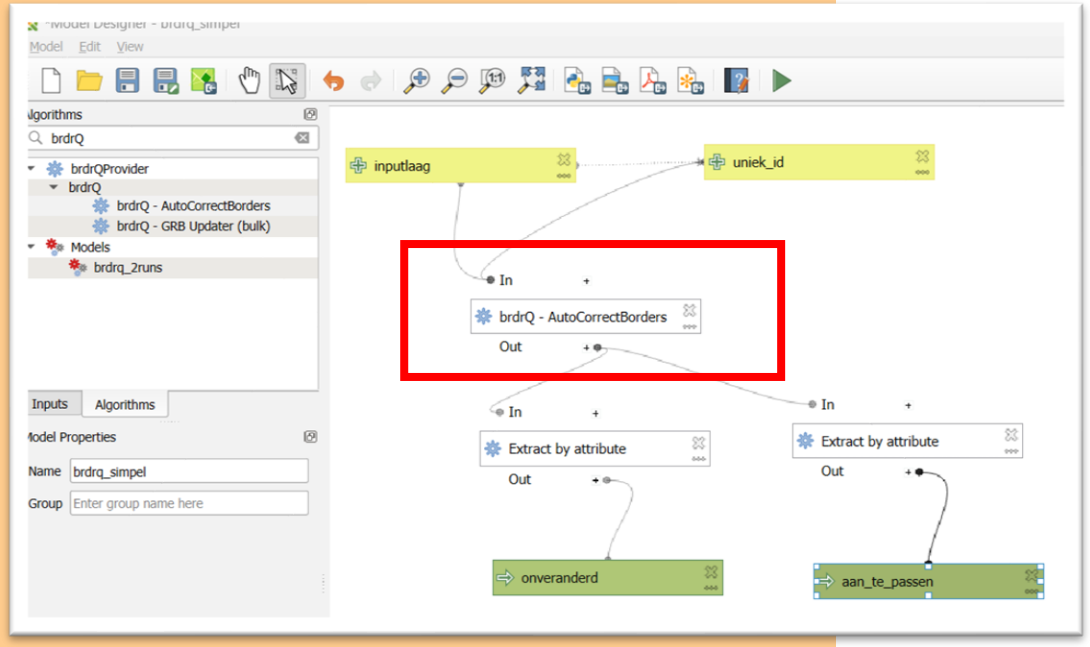


- QGIS plugin
- Visual UI to assist feature alignment
- Uses 'BRDR' library for processing
- Tools:

FeatureAligner	Feature-by-feature aligner with predictions
AutoCorrectBorders	Bulk alignment of dataset to a reference layer
GRB Updater	Bulk alignment to latest GRB (Flanders - Belgium)
Info	General info: version - documentation

https://github.com/OnroerendErfgoed/brdrQ

Architectural choices



- **BRDR (python) decoupled from brdrQ (QGIS)**
  - Flexibility
  - Pure Python scripting
  - Multiple integrations:
    - **QGIS**, *ESRI, FME, web service, …*

- **Why brdrQ (QGIS):**
  - Customer uses QGIS
  - Visualization & interaction with data
  - Free available – big range
  - Plugin Manager:
    - Good integration with python
  - Integrations in custom workflows:
    - Bulk alignment as QGIS Process
    - Availability in QGIS-Model Designer

HOW?

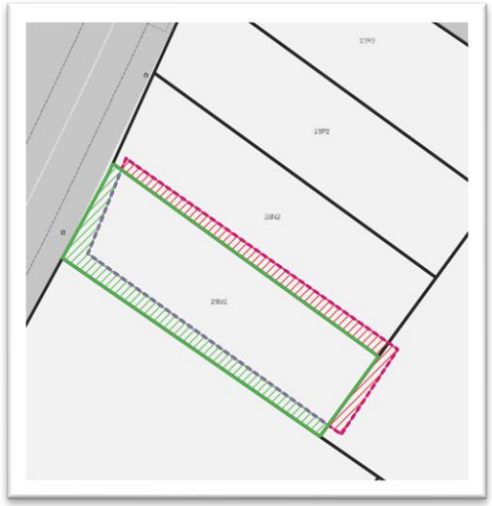
‘BRDR’ principles



INPUT DATA(SET)

BRDR

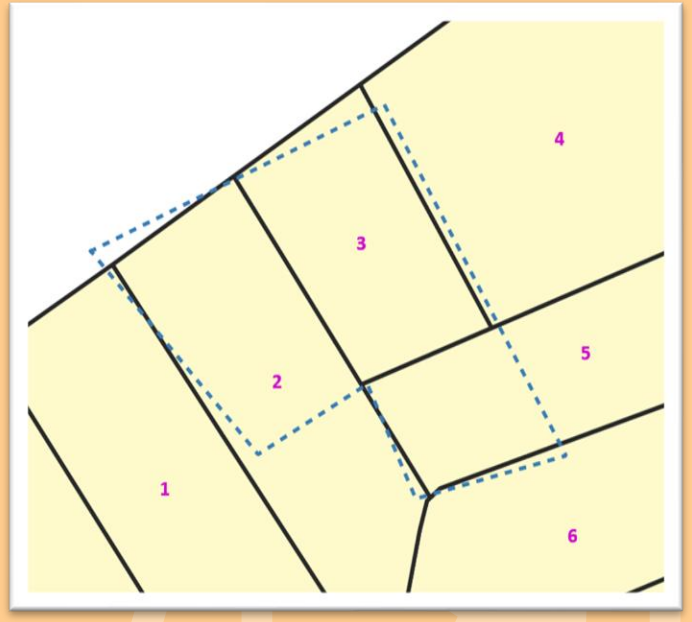
ALGORITHM

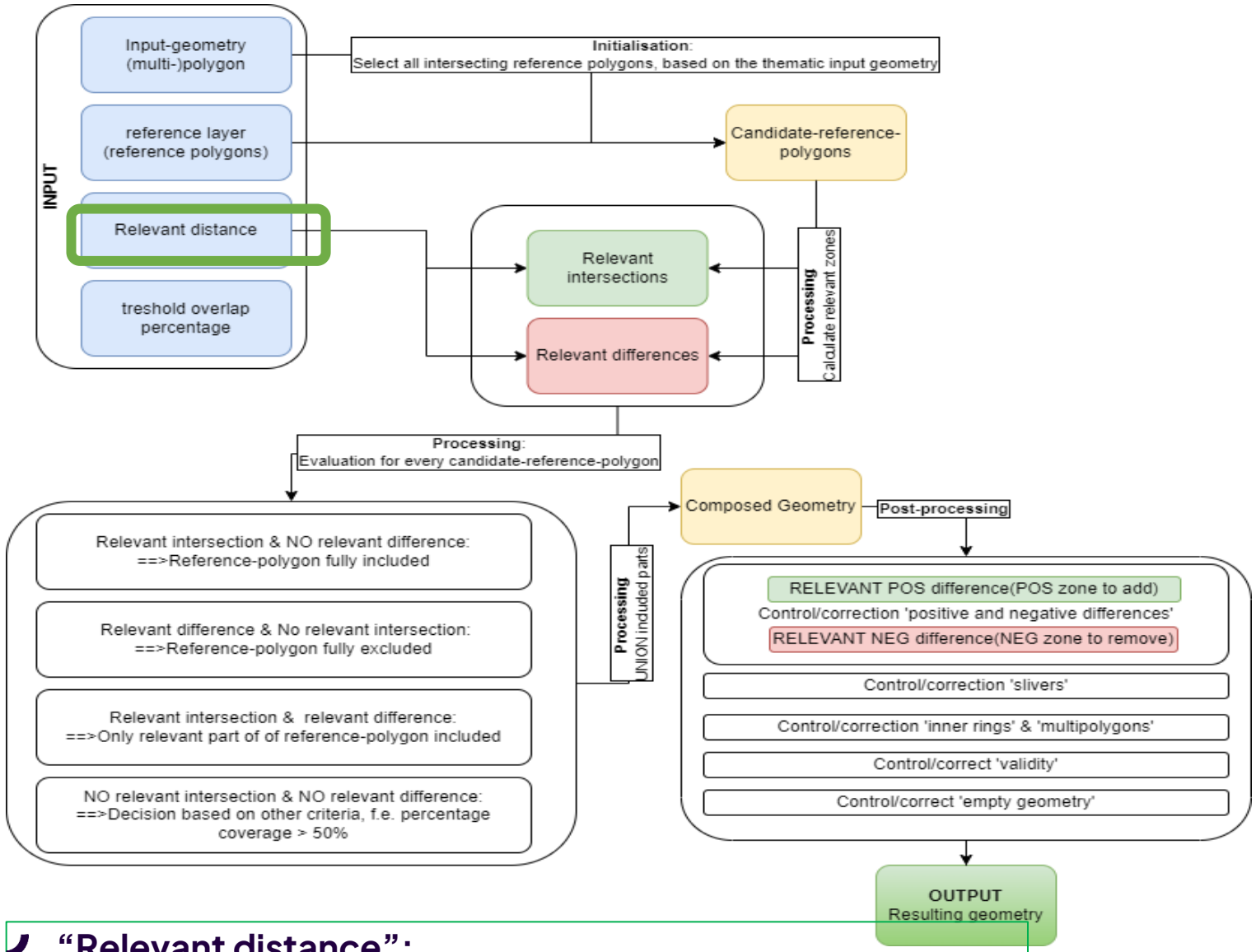


ALIGNED DATA(SET): PREDICTION(S) & DIFFERENCES

- **Conservation of shape:**
  - Use geometry of reference borders
- **Intentionality:**
  - Not every deviation is an error
  - intentional deviations should be retained

Alignment-algorithm ('BRDR')

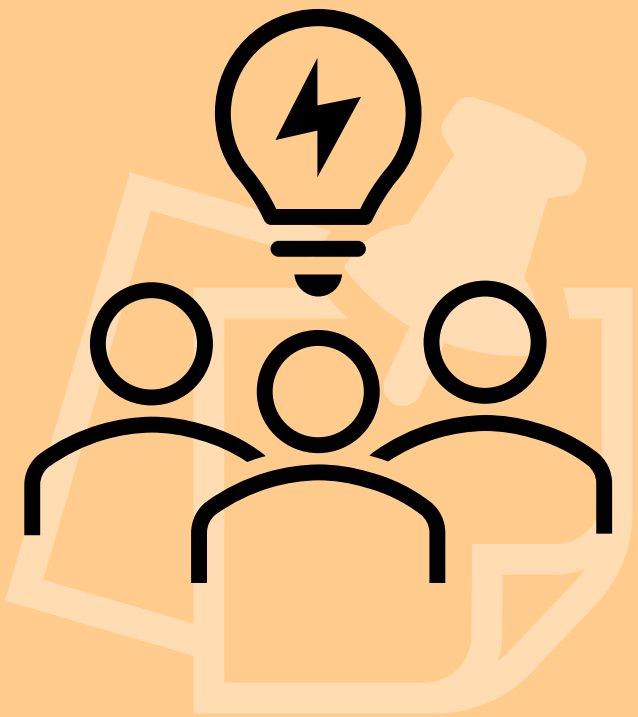




"Relevant distance":

- Distance used to determine which zones are included or excluded in the resulting geometry.
- ~Maximum distance that the resulting geometry may change.

Alignment-algorithm ('BRDR')

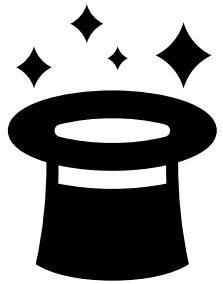


- "How to determine this 'relevant distance'?"



➔ 'BRDR' tries all !

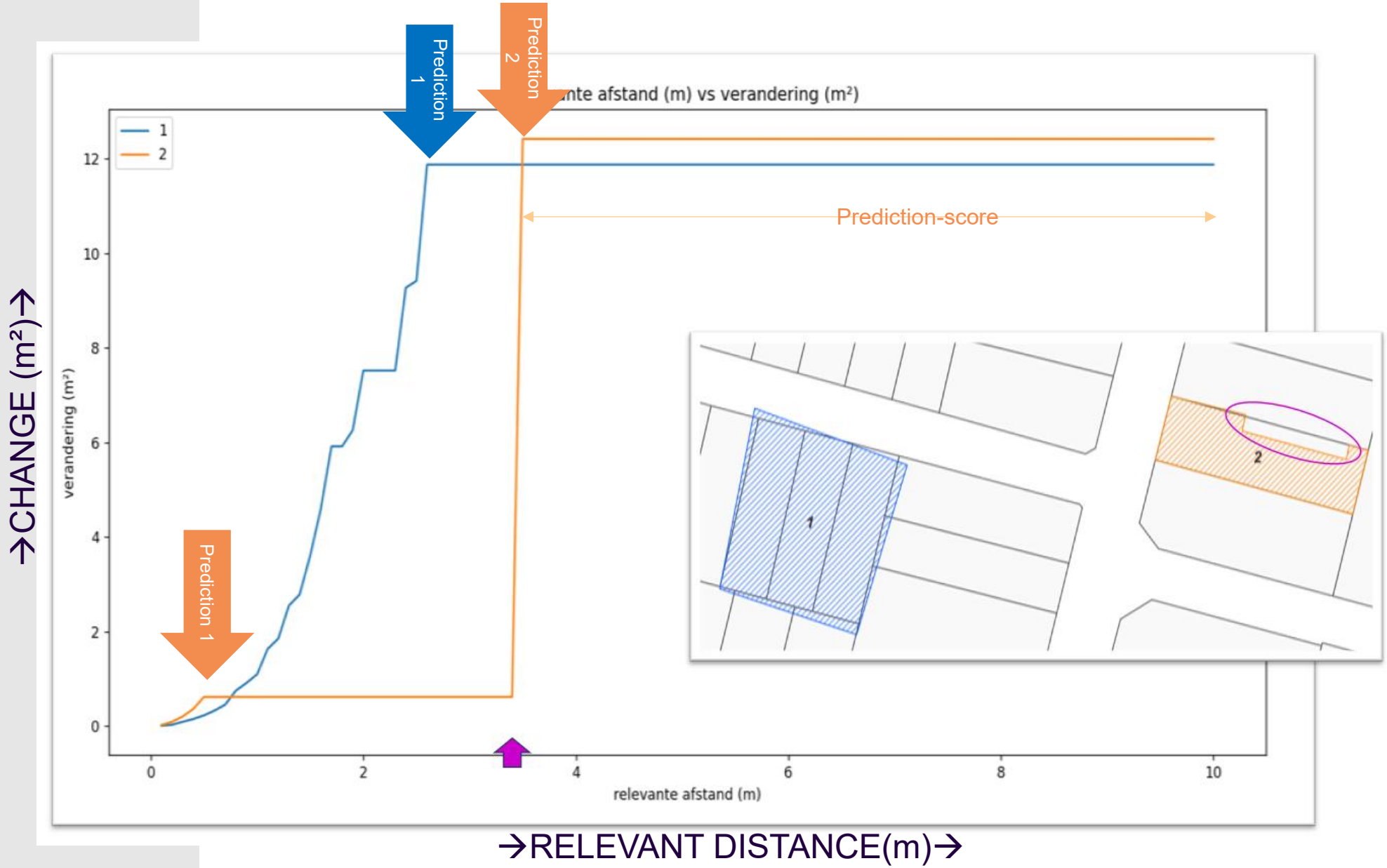
PREDICTIONS



Predictions

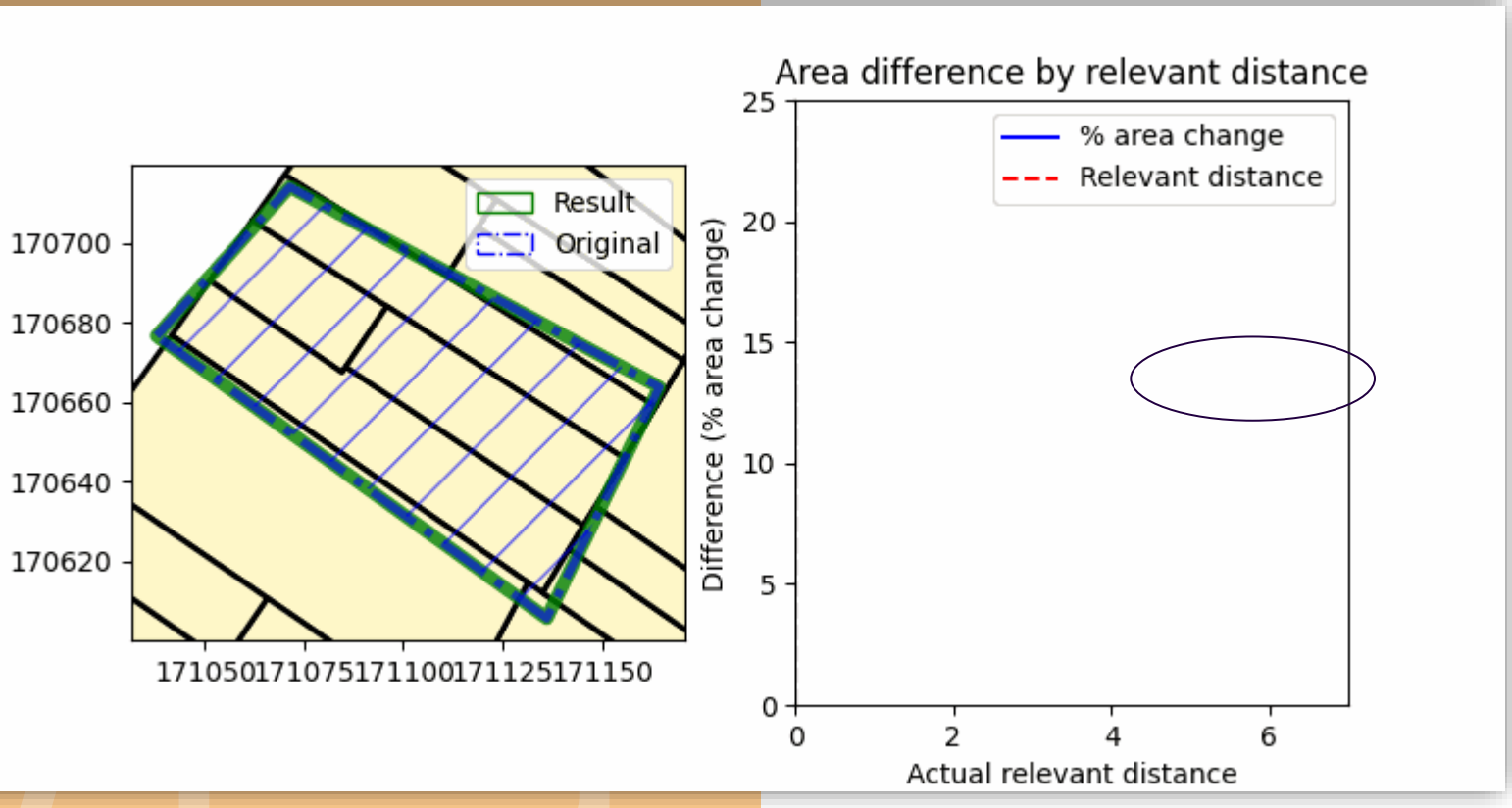
- **Parallel BRDR-algorithm execution with series of relevant distances**
  - 0m, 0.1m, 0.2m, …1m…1.9m, 2m
- **Measurement of change(m²) =** Difference of area between original & resulting geometry
- **PREDICTIONS=** 'Stable' results
- **PREDICTION SCORE=**

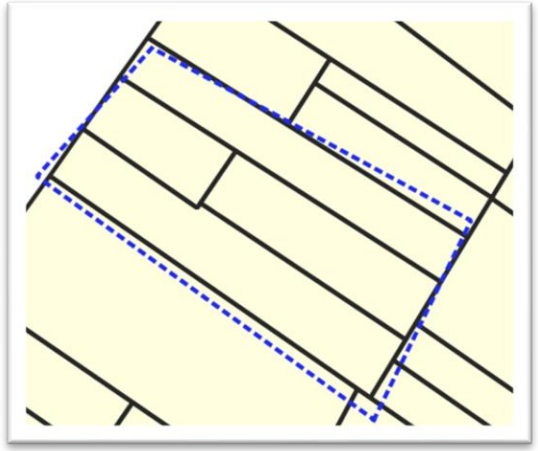
  Length of stability



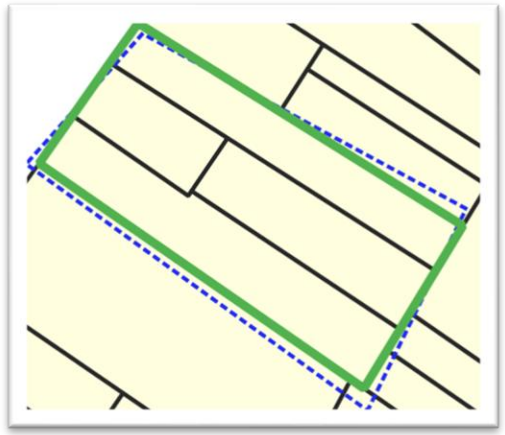
- **PREDICTIONS = "Stable result":**
  - A result that does not change when we augment the relevant distance

Predictions



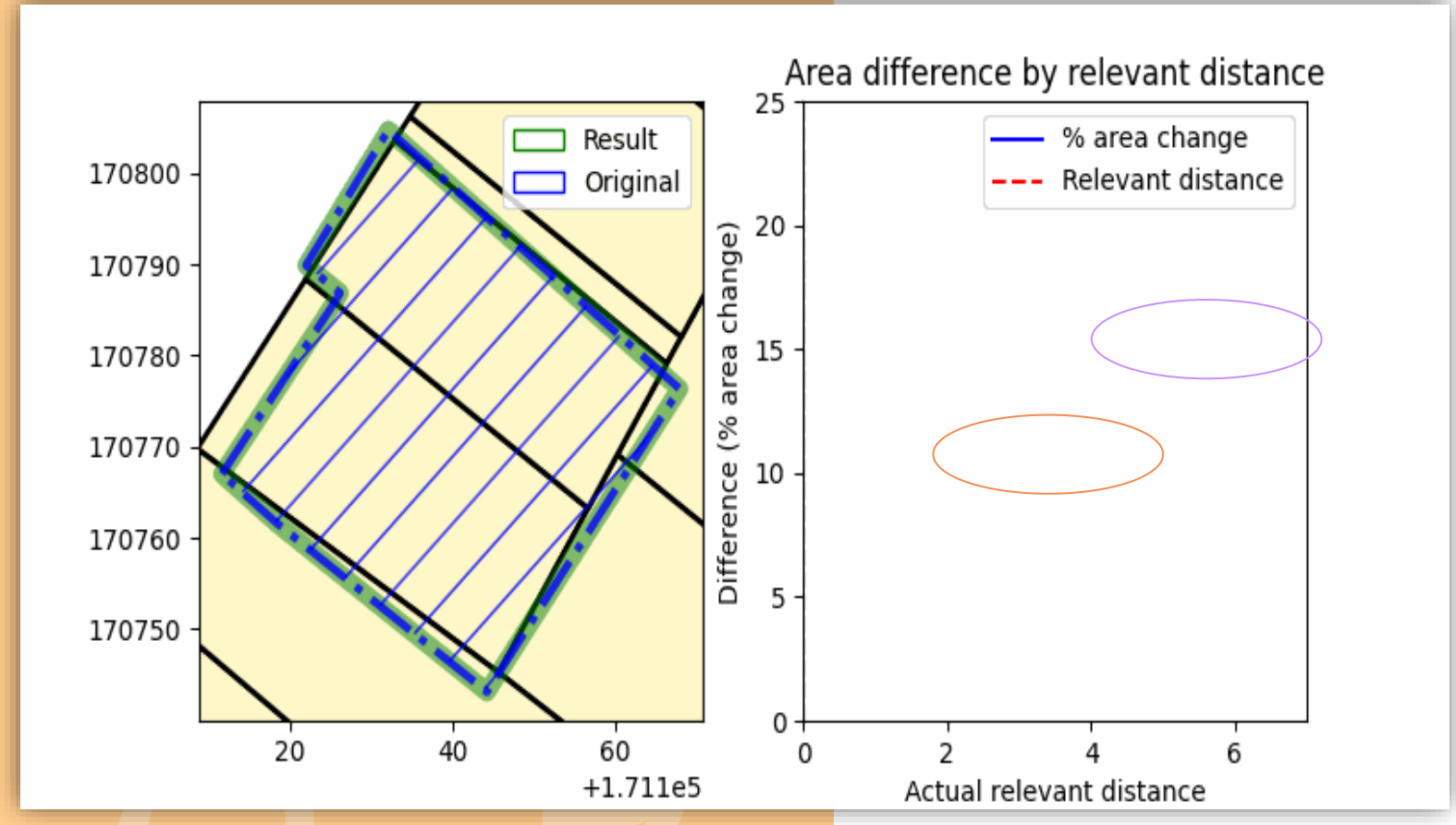


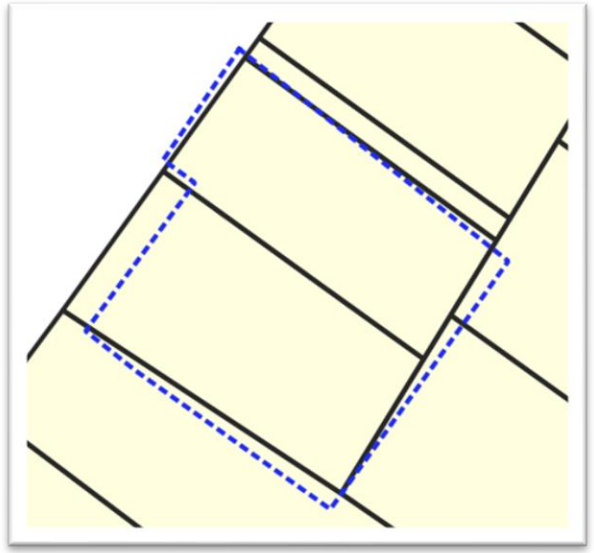
PREDICTION @ 4.8m



- **PREDICTION = "Stable result":** area doesn't change with increasing relevant distance

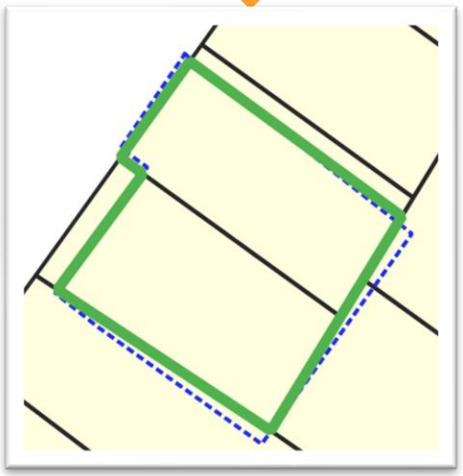
Predictions

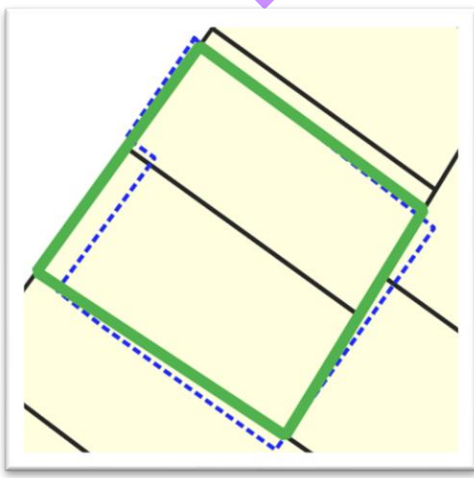




PREDICTION 1 @ 2.8m

PREDICTION 2 @4.4m





- **PREDICTION = "Stable result":** area doesn't change with increasing relevant distance

Other geometry types?



- **Initially: polygon-to-polygon** based on ‘relevant zones’
  - Relevant intersections & differences
  - Fast

- **Later on: support for all point-line-polygon (combinations)**
  - Based on graphs (networks)
  - Slower

Key strengths of BRDR

Intelligent, contextual and future-proof alignment of geometries with changing reference data.

1 Reference geometry is reused

BRDR aligns geometries by reusing existing reference boundaries instead of simply moving vertices within a tolerance.



- Original geometry
- BRDR aligned geometry (reusing reference boundaries)
- Reference geometry (e.g. cadastral/authoritative data)

2 Intentionality is respected

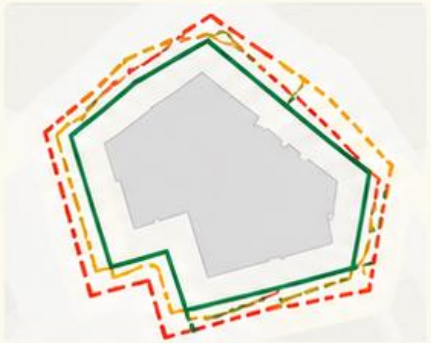
Not every deviation is an error. BRDR distinguishes between unintentional and intentional differences and keeps relevant thematic boundaries.



- Aligned geometry
- Intentional deviation (preserved)
- Reference geometry

3 Predictions enable multiple plausible results

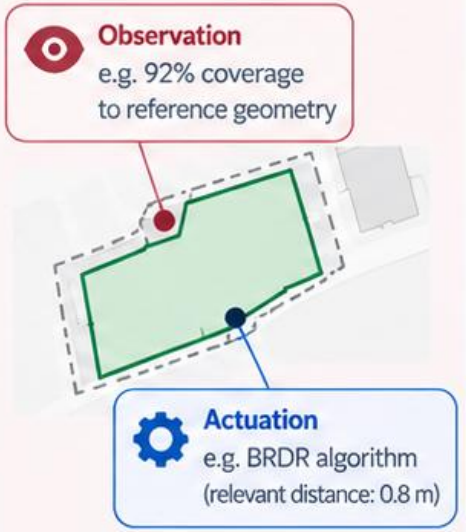
BRDR analyses different distances and identifies stable results. It can propose multiple candidate alignments instead of a single deterministic result.



- Candidate 1 (small distance)
- Candidate 2
- Candidate 3 (stable result)
- Original geometry

4 Context and semantics via brdr_metadata

BRDR records how a geometry relates to the reference, including observations and actuations, providing transparency and traceability.



5 Evaluation and realignment over time

BRDR makes it easy to evaluate results and realign geometries when reference data changes. This supports continuous maintenance and geodata lifecycle management.

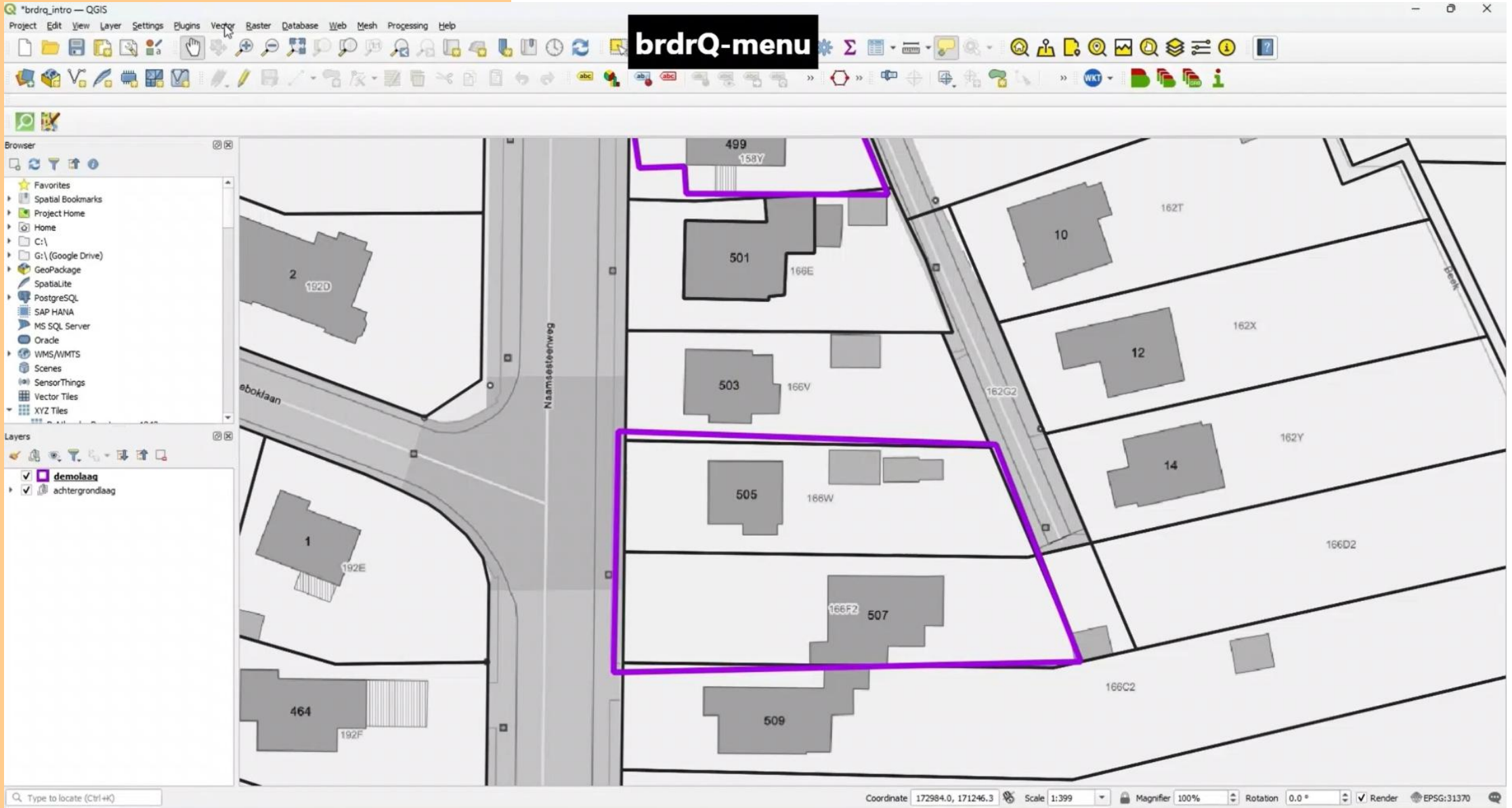


- Old reference
- New reference (update)
- BRDR realigned geometry

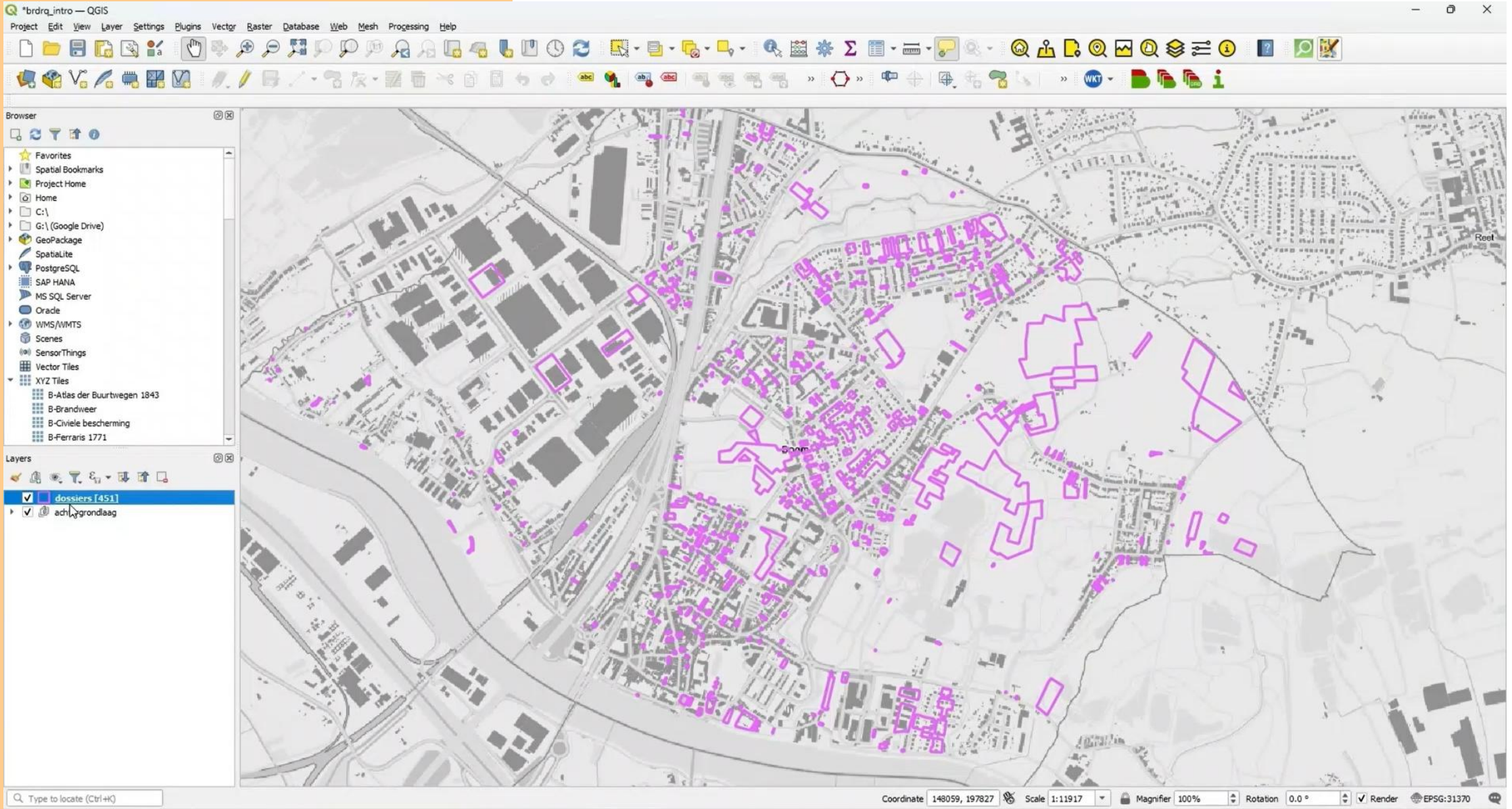
BRDR enables trustworthy geodata lifecycle management

Combining reference-based alignment, intentionality, predictions and contextual metadata to keep geometries consistent, transparent and manageable over time.

FeatureAligner



AutoCorrectBorders



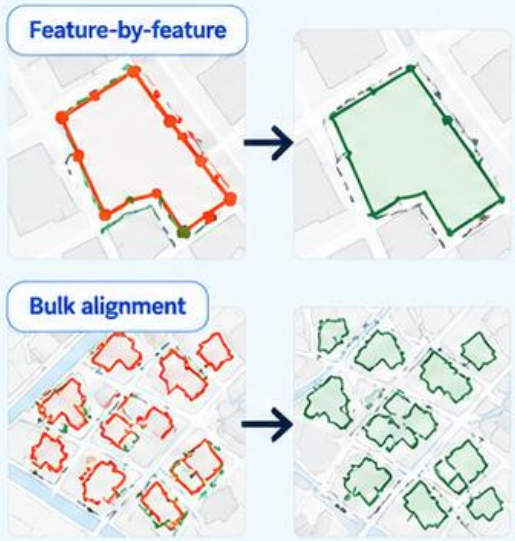
Use cases for BRDR

From day-to-day workflows to long-term geodata management and analysis.

1 Implementation in business processes, workflows and tooling

Integrate BRDR in your existing GIS workflows, tools and business processes.

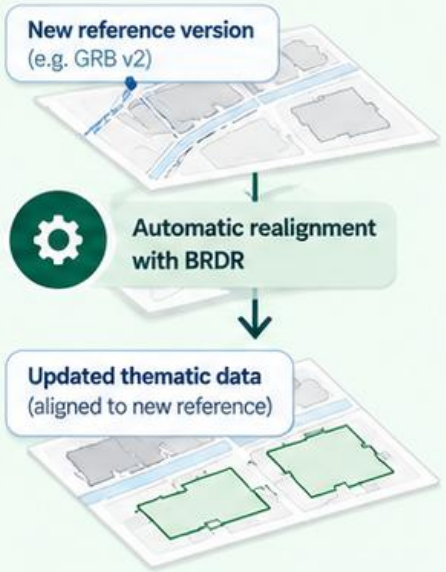
- Feature-by-feature alignment
- Bulk geodata alignment



2 Geodata lifecycle management

Automatically align and update your geodata to newer versions of reference data, and keep it consistent over time.

- Automatic actualizations to newer version of reference data



3 Version management

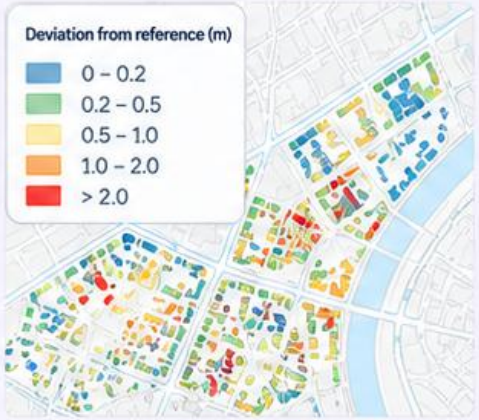
Visualize and analyse differences between versions of your geodata. Which objects changed, and which (not)?



4 Research & data analysis

Use BRDR to investigate patterns in deviation and change between thematic and reference boundaries.

- Maximum deviation
- Which reference layers best fit?
- In which areas or object types are deviations largest?
- Other spatial analyses ...



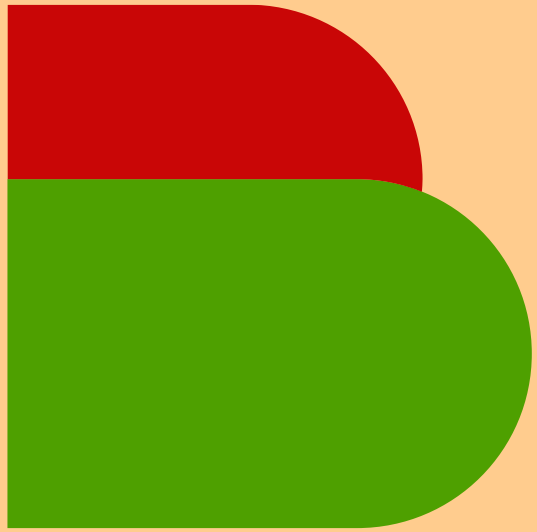
And more ...

- Data quality monitoring
- Compliance and reporting
- Integration with other datasets
- Decision support and scenario analysis
- Other use cases in your domain

BRDR supports a wide range of use cases

From operational alignment and updates to version management and in-depth spatial analysis – enabling reliable and future-proof geodata across its entire lifecycle.

Thanks! Questions?



Want to try?

- Python package 'brdr': https://github.com/OnroerendErfgoed/brdr
- QGIS – plugin brdrQ: https://github.com/OnroerendErfgoed/brdrQ
- Project page (dutch): https://docs.athumi.eu/vip/help/grenzen-verleggen
- Project report (dutch) : https://doi.org/10.55465/SXCW6218

Bugs?

- Brdr - bug? → https://github.com/OnroerendErfgoed/brdr/issues
- BrdrQ – bug?→https://github.com/OnroerendErfgoed/brdrQ/issues
- Idea?→ discussion forum: https://github.com/OnroerendErfgoed/brdr/discussions/

Ideas/Improvement/Discussion

- Discussion forum: https://github.com/OnroerendErfgoed/brdr/discussions/
- E-mail: karel.dieussaert@vlaanderen.be / Karel.dieussaert@geosquare.be

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